
Reference Free Grading of Reasoning Degrades on Problems Inside the Instella Pretraining Corpus

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Abstract

1 Automated evaluators are trained artifacts, so a verdict on a benchmark item may
2 depend on whether the evaluator encountered that item during training, yet the
3 corpora of models deployed as judges are closed and the question is normally
4 left open. Instella publishes its complete pretraining mixture and its stage two
5 component derives from the GSM8K training split rather than the test split, so
6 membership becomes a measured property fixed by exact thirteen gram containment
7 instead of an estimated one for judges built on that corpus. Grading 2017 candidate
8 solutions produced by Instella checkpoints, five judges spanning Instella 3B, Llama
9 3.1 8B and Qwen2.5 at 7, 14 and 32 billion parameters show a membership gap
10 in balanced accuracy that depends almost entirely on whether the gold answer
11 accompanies the prompt. Supplied with the reference the gap runs from -0.023
12 to -0.074 and shrinks as judges improve, reaching $-0.023 [-0.057, +0.006]$ at
13 14 billion parameters. Withheld, it runs to $-0.243 [-0.282, -0.202]$ and does
14 not close with scale, standing at $-0.191 [-0.238, -0.140]$ for the largest judge
15 tested. Decomposition locates the effect entirely in specificity: sensitivity gaps
16 span zero at every judge, at most $+0.011 [-0.005, +0.035]$, while specificity
17 falls by $-0.496 [-0.572, -0.416]$, meaning Llama 3.1 8B rejects 52.2 percent
18 of wrong solutions to unseen problems and 2.7 percent of wrong solutions to
19 seen ones. Familiarity produces credulity rather than confusion, which falsifies
20 an account based on comparison against a remembered answer since that predicts
21 a sensitivity loss. The effect survives a paraphrased judging prompt, -0.254
22 $[-0.293, -0.215]$, and weakens only where candidate errors are blunt enough
23 to catch without a reference, $-0.035 [-0.077, +0.007]$ for the strongest judge
24 on the weakest generator. Cohen’s kappa computed on identical verdicts reports
25 -0.485 where balanced accuracy reports -0.243 , an inflation traceable to a base
26 rate difference of 0.772 against 0.526 rather than to any change in agreement, so
27 chance corrected statistics should not size membership effects when arms differ
28 in base rate. The configuration in which the gap is largest, grading without a gold
29 answer, is exactly the configuration deployed whenever no gold answer exists.

1 Introduction

31 An LLM judge is asked to certify reasoning it may have been trained on. Whether that matters is an
32 empirical question that is normally unanswerable, because the pretraining corpora of models used
33 as judges are not published, so membership has to be estimated from proxies such as perplexity or
34 benchmark contamination heuristics. Instella [Liu et al., 2025] removes that obstacle. Its complete
35 pretraining mixture is released, and the stage two synthetic component is generated from the GSM8K
36 training split [Cobbe et al., 2021] and not from the test split. Membership therefore becomes a
37 property that can be measured on each item by exact thirteen gram containment rather than inferred,
38 which turns a question about contamination into a controlled contrast.

39 The design holds everything fixed except one binary variable. Identical benchmark items, identical
40 candidate solutions and identical judges are graded twice, once with the gold answer supplied and
41 once without. Reliability is scored as balanced accuracy against exact match ground truth, and the
42 quantity of interest is the seen minus unseen difference within each condition. Any property of the
43 items themselves, their difficulty, their phrasing or their length, enters both conditions equally and
44 cancels.

45 Three findings follow. Reliability loss on training corpus problems is small and closes with capability
46 when the reference is supplied, and large and persistent when it is withheld. The loss is a specificity
47 failure and not a sensitivity failure, so judges become credulous about familiar problems rather than
48 confused by them, which rules out the intuitive account in which a judge compares a candidate
49 against a remembered answer. Chance corrected agreement inflates the effect substantially on
50 identical verdicts, for reasons that follow from the marginals and not from the judging.

51 **Related work:** Judge reliability is normally studied through agreement with human raters [Zheng
52 et al., 2023]. Contamination work runs in parallel and asks whether a benchmark score is inflated
53 by training exposure, either by detecting membership from model behaviour [Shi et al., 2024], by
54 rephrasing items so that surface overlap fails [Yang et al., 2023], or by rebuilding a benchmark from
55 scratch to see how much of a reported score survives [Zhang et al., 2024]. That literature almost
56 always concerns the model under test rather than the model doing the grading, and memorisation is
57 quantified for generation rather than for judgement [Carlini et al., 2023]. The contribution here sits at
58 the join of the two. Membership is fixed by a published corpus rather than estimated from behaviour,
59 and the quantity measured is the evaluator’s reliability rather than the generator’s score.

60 2 Design

61 **Membership:** Containment is computed as the fraction of an item’s thirteen grams appearing in
62 a single corpus document, taking the maximum over documents. The stage two subset the model
63 actually consumed, `train_119K`, holds 119,014 rows. Calibration fixes the threshold: against the full
64 released pool of 1,367,882 rows, no GSM8K test item reaches containment 0.50 and the maximum
65 observed is 0.487, confirming the corpus is train derived and that the train against test partition is a
66 genuine exposure contrast rather than a difficulty contrast.

67 **Candidate solutions:** Instella checkpoints generate solutions to both arms under identical decoding.
68 Ground truth is exact match against the reference answer. Base rates differ between arms by
69 construction, 0.772 correct on the seen arm against 0.526 on the unseen, which is why balanced
70 accuracy rather than raw agreement is the primary statistic.

71 **Judging:** Each judge grades every solution twice. The reference given condition places the gold
72 answer in the prompt; the reference withheld condition is identical in every other respect. Verdicts
73 that fail to parse are excluded throughout, 3 of 2017 and 17 of 2017 for the largest judge.

74 **Inference:** Intervals are cluster bootstraps over parent problems with 4000 draws at seed 6198,
75 because several graded variants can descend from one problem and resampling rows rather than
76 problems would understate the interval.

77 3 Results

78 3.1 The gap depends on whether a reference is supplied

79 Table 1 gives the membership gap for every judge under both conditions. Supplied with the gold
80 answer, the gap is small and shrinks as judges improve, reaching -0.023 $[-0.057, +0.006]$ at 14
81 billion parameters, an interval containing zero. Withheld, the gap is three to eight times larger at
82 every judge and shows no tendency to close: the 32 billion parameter judge, whose reference free
83 balanced accuracy on the seen arm is the highest measured at 0.648, still carries a gap of -0.191
84 $[-0.238, -0.140]$.

85 A null or an effect from an incompetent judge would say nothing, so absolute performance is reported
86 alongside the gaps. Figure 2 shows both. Given the reference, the 7, 14 and 32 billion parameter

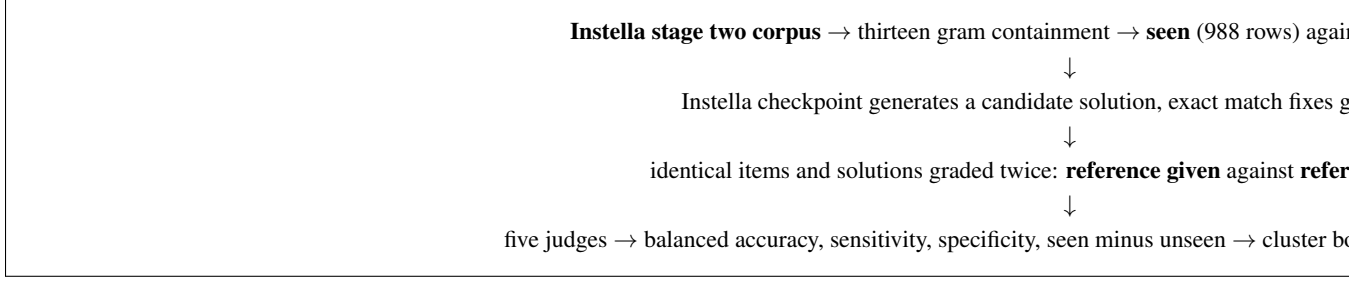


Figure 1: Membership is fixed by exact thirteen gram containment against the released Instella stage two corpus, giving 988 seen and 1029 unseen items. Items and candidate solutions are then frozen, so the only manipulated variable is whether the gold answer accompanies the judging prompt and every property of an item cancels across conditions. Verified membership holds for the Instella judge alone; for the others the partition is GSM8K train against test.

Table 1: Membership gap in balanced accuracy, seen minus unseen, grading instruct outputs. Intervals are cluster bootstraps over parent problems. An asterisk marks an interval excluding zero. The ratio column is the reference free gap divided by the control gap.

Judge	Reference given (control)	Reference withheld	Ratio
Instella 3B Instruct	−0.054 [−0.083, −0.026]*	−0.081 [−0.109, −0.054]*	1.5
Llama 3.1 8B Instruct	−0.074 [−0.116, −0.032]*	− 0.243 [−0.282, −0.202]*	3.3
Qwen2.5 7B Instruct	−0.036 [−0.068, −0.008]*	− 0.200 [−0.248, −0.149]*	5.6
Qwen2.5 14B Instruct	−0.023 [−0.057, +0.006]	− 0.184 [−0.231, −0.133]*	8.0
Qwen2.5 32B Instruct	−0.062 [−0.103, −0.022]*	− 0.191 [−0.238, −0.140]*	3.1

judges reach 0.888 to 0.973 balanced accuracy, which is the regime in which their reference free numbers become interpretable. The 3 billion parameter judge sits within 0.04 of chance on the seen arm in every configuration, including when handed the answer, so its rows describe a noise floor and the substantive claims rest on the larger three. The two control series converge as capability rises while the two reference free series stay apart, which is the dissociation stated as a picture.

3.2 The failure is specificity, not sensitivity

Balanced accuracy averages two rates that can move independently, and separating them identifies the mechanism. Figure 3 decomposes both conditions. Sensitivity, the rate of accepting correct solutions, carries no membership gap anywhere: +0.011 [−0.005, +0.035] for Llama 3.1 8B and −0.004 [−0.024, +0.016] for Qwen2.5 32B, both spanning zero. Specificity, the rate of rejecting wrong solutions, collapses: −0.496 [−0.572, −0.416] and −0.378 [−0.471, −0.281] respectively. In absolute terms Llama 3.1 8B rejects 52.2 percent of wrong solutions to unseen problems and 2.7 percent of wrong solutions to seen ones.

That asymmetry falsifies the account the design was originally built to test. A judge comparing a candidate against a remembered answer would mark correct solutions wrong whenever its memory were imperfect, producing a sensitivity loss. No such loss appears at any judge in either condition. What appears instead is credulity: familiarity with a problem raises the probability that a wrong solution to it is waved through.

3.3 Chance corrected agreement inflates the effect

Cohen’s kappa is the reflex choice for judge reliability and it misleads here. Figure 4 runs both statistics over one set of verdicts. With the reference supplied the two agree to three decimals, both reporting −0.074. With it withheld, kappa reports −0.485 where balanced accuracy reports −0.243, roughly double.

Nothing about agreement changed to produce that. Ground truth base rates are identical across the two conditions, 0.772 on the seen arm against 0.526 on the unseen, because the graded solutions do not change. What moves is the judge’s own verdict marginal, and kappa depends on both marginals as well as on agreement. An arm near one half carries more chance corrected headroom than a

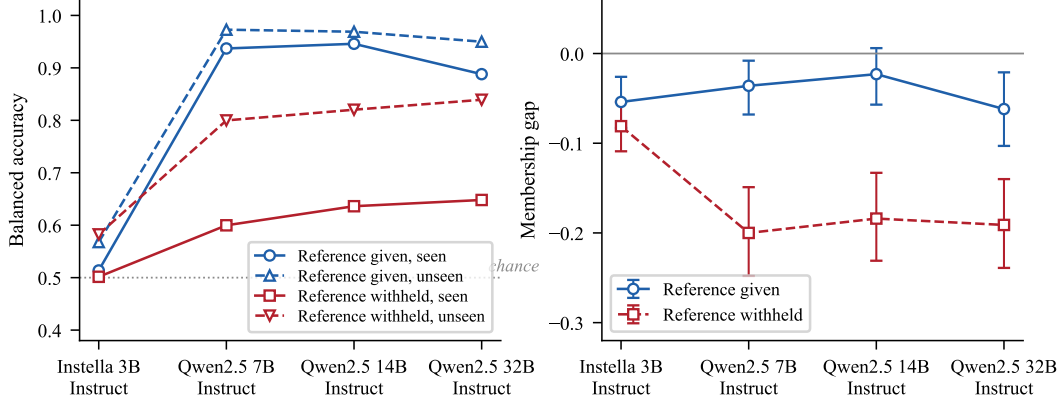


Figure 2: Left: absolute balanced accuracy for all four cells across the capability ladder. Right: the membership gap with cluster bootstrap intervals. Capability lifts every series and closes the control gap, leaving the reference free gap intact.

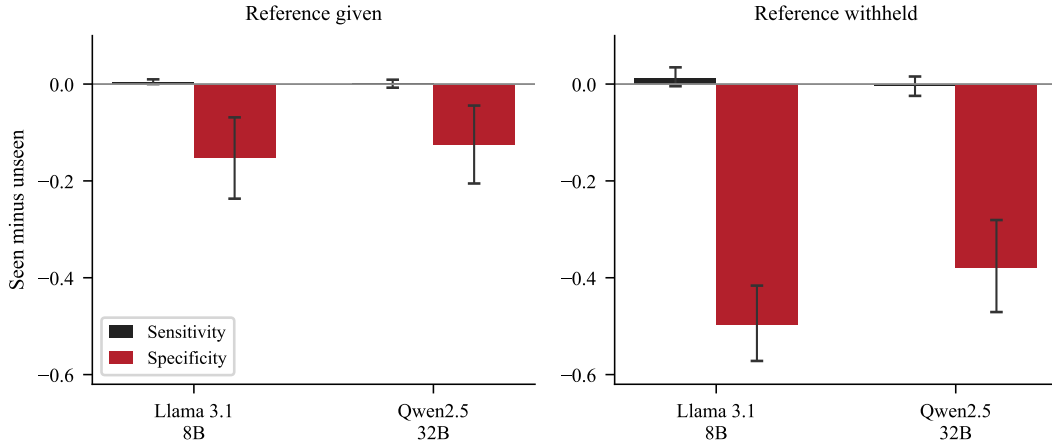


Figure 3: Sensitivity and specificity gaps, seen minus unseen, under both conditions. Sensitivity intervals contain zero everywhere. Specificity falls sharply once the reference is withheld, which is the entire effect.

114 skewed arm, so skew alone shifts the statistic. Feinstein and Cicchetti [Feinstein and Cicchetti, 1990]
 115 describe this as the first kappa paradox. The practical consequence for evaluator reporting is narrow
 116 and concrete: where arms differ in base rate, a chance corrected statistic will size a membership
 117 effect wrongly even when it points the right way.

118 3.4 Robustness

119 Two explanations compete with the account above and both are testable. Figure 4 addresses them.
 120 A paraphrased judging prompt, rewritten to preserve the task and change the wording, leaves the
 121 effect intact and slightly larger: -0.254 $[-0.293, -0.215]$ for Llama 3.1 8B against -0.243 with the
 122 primary prompt, and -0.219 $[-0.261, -0.172]$ for Qwen2.5 32B against -0.191 . Prompt wording
 123 does not drive it.

124 Varying the generator does matter, and in the direction the account predicts. Grading the weaker stage
 125 two checkpoint, whose errors are blunter, the strongest judge shows -0.035 $[-0.077, +0.007]$, an
 126 interval containing zero, while Llama 3.1 8B on the same target still shows -0.192 $[-0.232, -0.151]$.
 127 Errors coarse enough to catch without a reference leave nothing for familiarity to erode, so the effect
 128 has a ceiling in how gradeable the candidate solutions are.

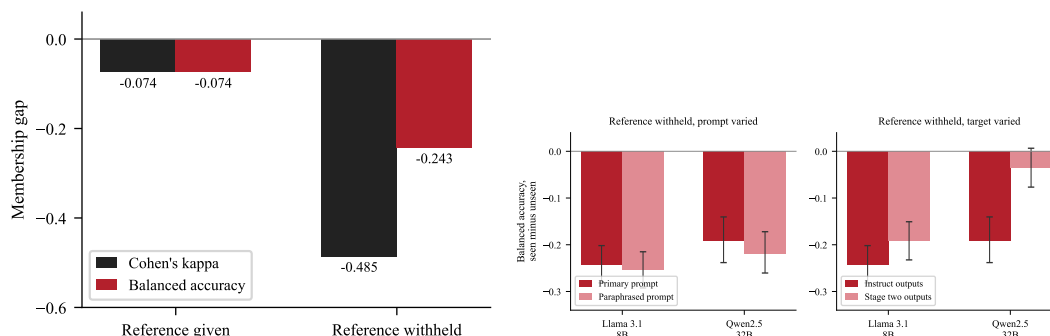


Figure 4: Left: two statistics on identical verdicts for Llama 3.1 8B. They agree under the control condition and diverge by a factor of two once the reference is withheld. Right: the reference free gap under a paraphrased judging prompt and against a weaker generator.

4 Implications for evaluator construct validity

A judge is deployed as an instrument measuring solution correctness. The results above say that instrument's specificity carries a term that depends on whether the graded item was in its training corpus, and that the term is nearly invisible whenever a gold answer is available. Benchmark practice makes this awkward, because the two conditions are not equally common. Reference free grading is used precisely where no gold answer exists, on open ended generation, on tasks scored by preference, and on new items for which references have not been written. Those are the settings in which the measured degradation is largest.

The size is operationally relevant rather than marginal. A judge rejecting 52.2 percent of wrong answers on unfamiliar problems and 2.7 percent on familiar ones is not applying one standard with noise. It is applying two standards, and which one applies is determined by a property of the judge's training data that the evaluation does not record and usually cannot observe. Reporting balanced accuracy split by sensitivity and specificity, rather than a single agreement number, makes the failure visible at low cost.

5 Limitations

Membership is verified for one judge, and it is not the judge carrying the result: Containment is measured against the Instella corpus, so the Instella judge has verified membership but sits at chance. The Qwen and Llama judges show the effect and their corpora are unknown, so for them the partition is GSM8K train against test rather than verified seen against unseen. That asymmetry is why the primary claim is stated as a within judge contrast between two conditions over identical items and identical candidate solutions, which holds whatever the arm label means for a given judge. Closing the gap to a membership claim requires a second open corpus model of comparable capability.

One family supplies the candidate solutions: Every graded output comes from an Instella checkpoint, and whether the dissociation reproduces on other generators is untested. The judges span three families.

The effect has a ceiling in solution quality: The stage two result above is measured on one generator pair and should be treated as an observation rather than a law.

One checkpoint is unexplained: Stage one outputs are wrong nearly everywhere and trivially gradeable in either condition, which accounts for that null. The supervised fine tuned checkpoint shows no gap in either condition despite output quality comparable to instruct, and the present data do not explain why.

Attribution is not established: No evidence links a verdict to a specific training document. Section 3.2 narrows the effect to a specificity loss, which is not attribution.

162 **A second, heavier exposure boundary exists and is untested:** Instella-MoE-16B-A3B, released
 163 after this study began, consumes the full released pool of 1,367,882 rows rather than the `train_119K`
 164 subset used here. Measured on identical items through identical extraction, 18.52 percent of GSM8K
 165 train parents reach containment 0.999 or above against that pool, compared with 9.65 percent against
 166 `train_119K`, so a judge built on the newer checkpoint would carry roughly double the verbatim
 167 exposure at the same items. Whether the specificity loss reported here scales with that heavier
 168 exposure is not measured and is left to future work.

169 6 Conclusion

170 Measured against membership fixed by containment rather than estimated, an LLM judge shows a
 171 small reliability loss on problems from its training corpus when the reference answer is supplied, and
 172 a large one when it is withheld. The gap under reference free grading runs to -0.243 in balanced
 173 accuracy and does not close with scale, while the same judges in the control condition converge
 174 toward zero as they improve. The loss is entirely a specificity failure, so familiarity buys credulity
 175 rather than confusion. Chance corrected agreement roughly doubles the apparent effect on identical
 176 verdicts and should not be used to size it where arms differ in base rate. The operational reading is
 177 specific: reference free grading is the configuration deployed wherever no gold answer exists, and it
 178 is the configuration in which the items most likely to sit in a judge’s training data are graded least
 179 reliably.

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218 **A Related work in full**

219 Zheng et al. [Zheng et al., 2023] introduce the LLM as judge paradigm at scale and document position
220 bias, verbosity bias and self preference, reporting agreement with human preference above 80 percent
221 for strong judges. Their audits characterize the judge and the prompt; membership of the graded item
222 in the judge’s corpus is outside reach for the closed models they study, which is the gap the present
223 design addresses for one open family.

224 On the detection side, containment based overlap rules originate with Brown et al. [Brown et al.,
225 2020], and membership inference from model confidence is developed by Carlini et al. [Carlini et al.,
226 2023] and Shi et al. [Shi et al., 2024]. Yang et al. [Yang et al., 2023] show that rephrasing defeats
227 overlap based detection outright, which is why a study that needs a trustworthy treatment label cannot
228 rely on an estimated one.

229 Two lines of work establish the value of released intermediate checkpoints. Biderman et al. [Biderman
230 et al., 2023b] release the Pythia suite specifically to make training dynamics analyzable, and Biderman
231 et al. [Biderman et al., 2023a] use such checkpoints to study when memorization emerges during
232 training. The present work applies the same logic to an evaluator rather than to a generator.

233 **B Containment method and calibration**

234 Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13
235 consecutive tokens. Containment against a corpus is the fraction of an item’s windows found in it,
236 computed in one streaming pass with an inverted index from window to source item, so cost is linear
237 in corpus size and independent of how many item sets are indexed together. Because the corpus is
238 generated only from the GSM8K training split, every test item is a verified negative: the any match
239 rule flags 3 of 1000, a rate of 0.3 percent, and the adopted threshold flags none.

240 Identifier namespacing is load bearing. GSM8K train and test share an index scheme, so loading
241 both without namespacing collides items and produces containment values above unity, which is
242 impossible for a fraction. Anyone reproducing this design will meet the same trap.

243 **C Reproducibility controls**

244 Several controls are load bearing rather than incidental. The transformers version is pinned exactly,
245 since a version range places different checkpoints on different minor versions and that difference
246 enters the estimate as though it were a model difference. Batch size is fixed across all generation,
247 because greedy decoding in reduced precision is sensitive to padding and a batch size change alters
248 tie breaking. Variants are seeded per pair of item and perturbation type, so variant text is stable
249 regardless of call order and generations stay valid when the item set changes. Every generation batch
250 is flushed to disk and resume is keyed on item identifier, so an interrupted run continues instead of
251 restarting.

252 All generation records, containment values, judge verdicts and analysis code are released for audit at
253 a public repository and dataset mirror, both anonymized for review. Random seeds are fixed across
254 arm selection, variant construction and every bootstrap. An assistant helped implement generation,
255 containment and analysis code under the direction of the authors, who verified every reported number
256 against the underlying verdict files; the assistant did not design the conditions or choose which results
257 to report.